Name: ____________________ **Roll Number:** ____________________

Question:	1	2	3	4	5	6	Total
Points:	0	5	5	7	7	6	30
Score:							

1. Every question perfectly solved below will add an additional $1\frac{1}{2}$ points to your total.

2. (5 points) **Regression with Corrupted Measurements**

   Recall the regression problem from the first assignment in which we observed a corrupted vector $\underline{y}$ derived from $\underline{y}_{\text{orig}} = A\underline{x}_{\text{true}} + \underline{\epsilon}$ by flipping the sign of entries of $\underline{y}_{\text{orig}}$ with probability p. As in the assignment, the entries in $\underline{x}_{\text{true}}$, A, and $\underline{\epsilon}$ are generated i.i.d $\mathcal{N}(0, 1)$.

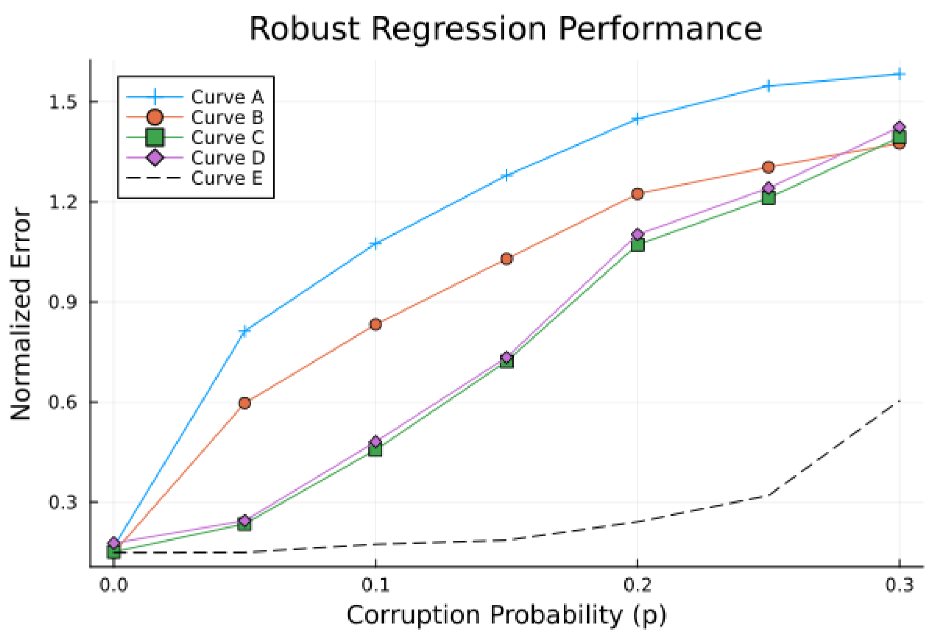
   We attempt to recover $\underline{x}_{\text{true}}$ by solving the minimization problem:

   $$\hat{\underline{x}} = \arg\min_{\underline{x}} J(A\underline{x} - \underline{y})$$

   where $J(\cdot)$ is a cost function acting on the residual vector $\underline{r} = A\underline{x} - \underline{y}$. We test five different approaches:

   - **Least Squares:** $J(\underline{r}) = \|\underline{r}\|_2^2 = \sum r_i^2$
   - **Least Absolute Deviations:** $J(\underline{r}) = \|\underline{r}\|_1 = \sum |r_i|$
   - **Minimax:** $J(\underline{r}) = \|\underline{r}\|_\infty = \max_i |r_i|$
   - **Huber Loss:** $J(\underline{r}) = \sum \phi_{\text{huber}}(r_i)$
   - **Prescient Solution:** The oracle solution that knows the positions of the corrupted measurements and discards those measurements.

   The plot below shows the normalized error $\mathbb{E}\left[\frac{\|\hat{\underline{x}} - \underline{x}^{\text{true}}\|_2}{\|\underline{x}^{\text{true}}\|_2}\right]$, averaged over 50 trials as a function of the corruption probability $p \in [0, 0.3]$.



Identify which **Curve (A–E)** are likely to correspond to each of the five approaches. Provide a brief justification for your choice for each curve based on the properties of the cost function. If two methods produce indistinguishable curves, you may group them together in your answer and explain why they behave similarly in this specific context.

3. (5 points) **Moreau Envelopes**

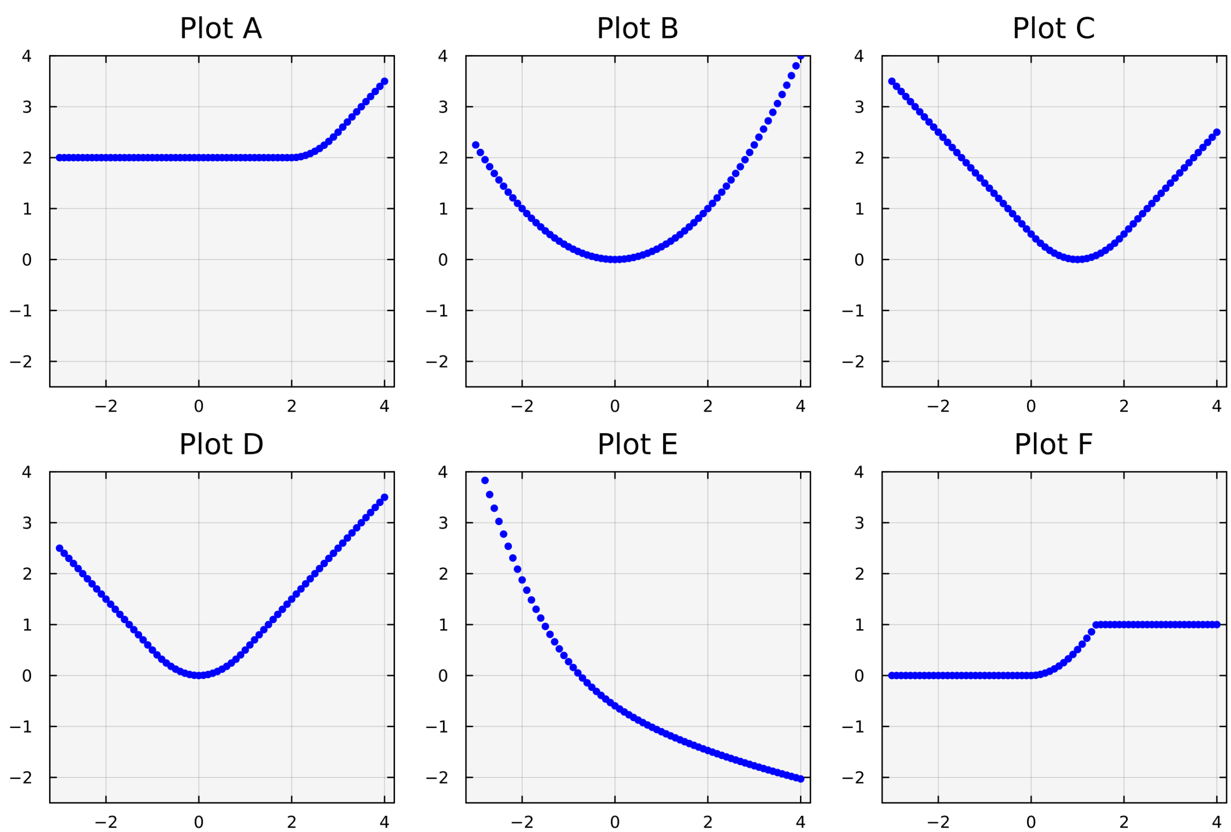
The Moreau envelope (with parameter $\lambda = 1$) of a function $f : \mathbb{R} \to \mathbb{R} \cup \{+\infty\}$ is defined as:

$$M_f(y) = \inf_x \left(f(x) + \frac{1}{2}(x - y)^2 \right)$$

Consider the following five single-variable functions:

- **Function 1:** $f(x) = |x - 1|$
- **Function 2:** $f(x) = \max\{x, 2\}$
- **Function 3:** $f(x) = \begin{cases} 0 & x < 0 \\ 1 & x \geq 0 \end{cases}$
- **Function 4:** $f(x) = \begin{cases} -\sqrt{x} & x \geq 0 \\ +\infty & x < 0 \end{cases}$
- **Function 5:** $f(x) = \frac{1}{2}x^2$

The figure below displays 6 plots (labeled A–F) of computed Moreau envelopes $M_f(y)$. Five of these plots correspond to the functions above.



Match each **Function (1–5)** to its corresponding **Plot (A–F)**, and justify your choice.

4. *Projection, Constrained Least Squares, and Pareto Optimality*

Consider the following sequence of optimization problems where we enforce an equality constraint on the norm of the solution. For parts (a), (b), (c):

- Provide a closed form solution to find x.
- Derive the optimal value of the objective function in terms of the given parameters (e.g., z, y, γ, Σ).
- Describe the computational complexity.

(a) ($1\frac{1}{2}$ points) *Spherical Projection:* Let $z \in \mathbb{R}^n$ be a given vector and $\gamma > 0$ be a scalar. Solve (note the problem asks to find projection on the surface of the sphere)

$$\min_x \|x - z\|_2 \quad \text{subject to} \quad \|x\|_2 = \gamma$$

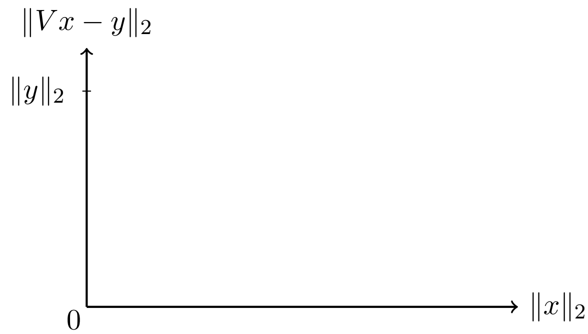
(b) ($1\frac{1}{2}$ points) *Restricted Identity Regression with constraints:* Let $\Sigma \in \mathbb{R}^{m\times n}$ (where $m > n$) be a matrix with 1s on the main diagonal and 0s elsewhere. Let $y \in \mathbb{R}^m$.Solve

$$\min_x \|\Sigma x - y\|_2 \quad \text{subject to} \quad \|x\|_2 = \gamma$$

(c) (2 points) *Orthogonal Basis Regression with constraints:* Let $V \in \mathbb{R}^{m \times n}$ $(m > n)$ be a tall matrix with orthonormal columns ($V^T V = I_n$). Let $y \in \mathbb{R}^m$. Solve:

$$\min_x \|Vx - y\|_2 \quad \text{subject to} \quad \|x\|_2 = \gamma$$

(d) (2 points) *Pareto Optimal Curve:* Consider the multi-objective minimization problem with objectives $f_1(x) = \|Vx - y\|_2$ and $f_2(x) = \|x\|_2$. Using your result from Part (c), Sketch the Pareto optimal curve in the 2D plane below (x-axis: $\|x\|_2$, y-axis: $\|Vx - y\|_2$). Clearly mark the intercepts and the global unconstrained minimum. When does this curve touch the horizontal axis ? Comment on the convexity and monotonicity of the curve.



5. *Regression with norm constraints:*

(a) (4 points) Let $\Sigma \in \mathbb{R}^{m \times n}$ be a diagonal matrix with arbitrary non-negative diagonal entries $\sigma_1 \geq \cdots \geq \sigma_n \geq 0$. Find the vector $x \in \mathbb{R}^n$ that solves:

$$\min_x \|\Sigma x - y\|_2 \quad \text{subject to} \quad \|x\|_2 \leq \gamma$$

Hint: Use KKT conditions

(b) (3 points) *General Constrained Least Squares:* Let $A \in \mathbb{R}^{m \times n}$ be an arbitrary tall matrix ($m \geq n$) and $b \in \mathbb{R}^m$. Find the vector $x \in \mathbb{R}^n$ that solves:

$$\min_x \|Ax - b\|_2^2 \quad \text{subject to} \quad \|x\|_2 \leq \gamma$$

6. *Conjugate and Envelope of the Max Function*

(a) (3 points) Let $f : \mathbb{R}^n \to \mathbb{R}$ be defined by the maximum function: $f(x) = \max\{x_1, x_2, \ldots, x_n\}$. Derive the convex conjugate function $f^*(z)$, defined as: $f^*(z) = \sup_{x \in \mathbb{R}^n} \left(z^T x - f(x)\right)$. You may find it helpful to first consider the following cases (a) when one of the entries in z is negative, and (b) when all the entries in x are equal.

(b) (3 points) Let $M_{\mu f}(x) = \min_y \left(f(y) + \frac{1}{2\mu}\|x - y\|_2^2 \right)$ be the Moreau envelope of f with parameter $\mu > 0$. Express the gradient $\nabla M_{\mu f}(x)$ in terms of a Euclidean projection onto an appropriate set S, and explicitly identify the set S.